

BREADFRUIT

Layers

- Sewer Lines
- Reservoirs
- Overhead Tanks
- Treatment Plants
- Pumping Stations
- Manholes
- House Connections
- Service Zones

Legend

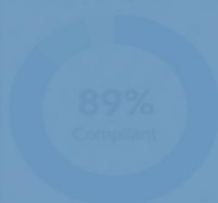
- Water System
- Sewer System
- Treatment Plant
- Service Area
- House Connection

JUNE 2026

INCEPTION REPORT

Regulatory Roadmap
and Partner
Coordination Dashboard

Water Quality (Sampling Points)



Good	69%
Fair	18%
Poor	3%

Asset Condition

+977 982-1635060

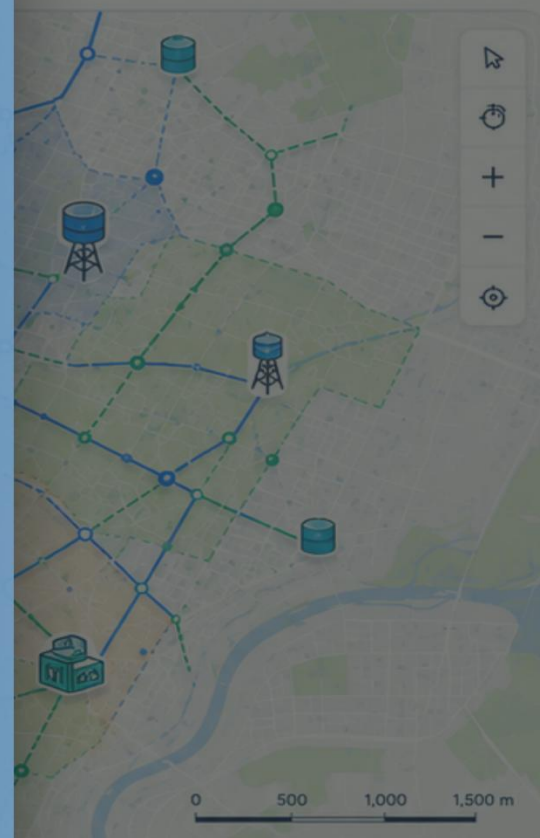
www.breadfruit.com.np

breadfruitpvtltd28@gmail.com

Good	72%
Fair	18%
Poor	3%

May 20 – May 26, 2024

Filter



0 500 1,000 1,500 m

Water Supplied (MLD)

52.4

MLD

+6% last week

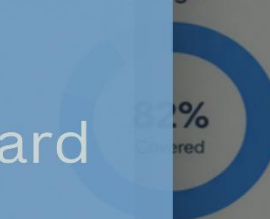


Sanitation Access

78%

+4% from last month

Service Coverage



Covered	82%
Partial	12%
Unserved	6%

Sanitation Access

Sewer Connection	68%
Septic Tank	20%
Open Defecation	12%

Alerts (7)

Low Pressure	Zone in Sector 7	2h ago
Water Quality	Alert in Zone 8	5h ago
Sewer Overflow	reported in Ward 12	1d ago

[View all alerts](#)



Executive Summary

This Inception Report has been prepared by BreadFruit Pvt Ltd in fulfilment of the first contractual milestone under the agreement entered into with the Wateraid Nepal for the design, development, and deployment of the Regulatory Roadmap and Partner Coordination Dashboard (RRPCD). The report presents the Consultant's confirmed understanding of the assignment, the proposed technical methodology, system architecture, work plan, and governance arrangements that will govern delivery of this engagement.

The institutional context for this assignment is DWSSM's formal establishment of the Service Regulation and Information Management Section (SRIMS) under the Service Regulation and Coordination Division, following the Government of Nepal's approval of a revised organisational structure for the Ministry of Water Supply in 2081. The enactment of the Water Supply and Sanitation Water Supply and Sanitation Regulations 2081 has created a substantive regulatory mandate — encompassing operator licensing, KPI-based performance monitoring, tariff oversight, and enforcement — that DWSSM/SRIMS is now legally required to operationalise. At present, performance data on Nepal's 87 registered Water Supply and Sanitation Operators (WSSOs) is dispersed across spreadsheets, sectoral reports, and institutional records, and no dedicated digital platform exists to support regulatory oversight as a distinct institutional function.

The RRPCD is conceived as a purpose-built, transitional digital platform that consolidates regulatory milestones, operator KPI performance data, and Development Partner coordination into a single, user-friendly web interface. It is architecturally independent of the existing National WASH MIS (NWASH-MIS), which serves local government service delivery monitoring and is not configured for regulatory data logic, while remaining designed for future integration into the national WASH information ecosystem as regulatory functions mature.

The platform will be structured around four core modules: a Dashboard Overview with high-level KPI summaries; an animated, interactive Regulatory Roadmap; an Operator Performance Module for the 87 WSSOs; and a Development Partner Coordination Module mapping DP contributions against regulatory implementation phases. An administrative backend will enable non-technical SRIMS staff to update data through a simple web form interface.

The Consultant proposes an eight-phase delivery methodology spanning requirement collection, data collection, system design, development, testing and quality assurance, deployment, training and handover, and capacity building and knowledge transfer — all within the 90-day assignment duration. The final phase concludes with the delivery of a production-deployed dashboard, a User Manual, and Technical Documentation, with a formal handover and knowledge transfer session for designated SRIMS staff.

This report is submitted for the review and written approval of WaterAid with recommendations from DWSSM/SRIMS. Upon approval, the Consultant will proceed immediately to the Requirement Collection phase in accordance with the work plan presented herein.

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1. Introduction

1.1 Background

The Department of Water Supply and Sewerage Management (DWSSM) is the principal government body responsible for planning, regulating, and monitoring water supply and sanitation services across Nepal. In 2081, the Government of Nepal approved a revised organisational structure for the Ministry of Water Supply that formally established service regulation as a core institutional function. This restructuring created the Service Regulation and Coordination Division and, within it, the Service Regulation and Information Management Section (SRIMS) — a unit with a specific mandate to build regulatory oversight capacity for the water and sanitation sector.

Concurrent with this institutional reform, the Water Supply and Sanitation Water Supply and Sanitation Regulations 2081 were enacted as the principal secondary legislation under the Water Supply and Sanitation Act 2079. These Regulations establish a comprehensive regulatory framework governing Nepal's 87 registered Water Supply and Sanitation Operators (WSSOs), including provisions on performance-based Key Performance Indicators (KPIs), licensing requirements, monitoring and reporting frameworks, tariff regulation, and consumer protection. Taken together, the Act and Regulations create a legally binding mandate for systematic regulatory oversight that demands purpose-built information management systems.

The National WASH MIS (NWASH-MIS) already exists as the sector's central data platform. However, it is designed to serve local government-level service delivery monitoring and is not architected to accommodate the distinct requirements of regulatory oversight. Integrating regulatory functions directly into NWASH-MIS at this early stage carries significant risks to the integrity and usability of the existing system. The long-term vision is eventual harmonisation of both systems; however, this transition must be managed carefully and incrementally.

It is against this backdrop that DWSSM/SRIMS has initiated the development of the Regulatory Roadmap and Partner Coordination Dashboard (RRPCD) — a dedicated, transitional digital platform that will consolidate regulatory milestones, operator performance data, and Development Partner coordination into a coherent institutional information management tool.

1.2 Purpose of This Report

This Inception Report serves as the formal first deliverable of the RRPCD engagement and fulfils the following purposes:

- To document the Consultant's confirmed understanding of the project objectives, scope, and expected outcomes following inception phase activities.
- To present the proposed technical methodology and phased approach for system design, development, and deployment.
- To describe the proposed system architecture, technical stack, and module design that will govern the development process.
- To define the work plan and milestones for project delivery within the 90-day assignment duration.
- To identify key risks, assumptions, and dependencies relevant to this engagement.
- To serve as the baseline reference document against which all subsequent project activities, deliverables, and decisions will be measured.

1.3 Project Scope

The scope of the RRPCD assignment encompasses the following:

- Design and development of a responsive, browser-based web dashboard for use by DWSSM/SRIMS staff and designated external stakeholders.
- Development of four core functional modules: Dashboard Overview, Regulatory Roadmap, Operator Performance Module, and Development Partner Coordination Module.
- Development of a backend data structure capable of ingesting operator and KPI data from the existing SRIMS master dataset in Excel format.
- Development of a simple administrative web form enabling non-technical SRIMS staff to update dashboard content.
- Delivery of a User Manual and Technical Documentation package.

The following items are explicitly outside scope unless otherwise agreed in writing:

- Full GIS/spatial visualisation integration (a placeholder integration point will be included; full GIS development is scoped for a future phase).
- Direct integration with NWASH-MIS during this phase (the architecture will be designed for future compatibility).
- Development of a mobile application or offline-capable data collection tool.
- Procurement of server infrastructure or cloud hosting on behalf of the client.
- Long-term operational maintenance beyond the agreed handover period.

1.4 Project Objectives

The RRPCD assignment is designed to achieve the following objectives:

1. To design and develop a functional, web-based dashboard that enables DWSSM/SRIMS to visually monitor the regulatory implementation roadmap, track operator KPIs, and coordinate with Development Partners in a transparent, data-driven manner.
2. To develop an interactive, visually coherent dashboard aligned with DWSSM/SRIMS institutional branding and operational workflows.
3. To integrate a structured data layer reflecting the performance of 87 registered WSSOs against defined KPIs, drawing from the existing SRIMS master dataset.
4. To create a Development Partner coordination module that maps financial and technical commitments against regulatory implementation milestones.
5. To ensure the platform is manageable by non-technical SRIMS staff, with administrative controls for content updates.
6. To deliver a maintainable, well-documented codebase suitable for iterative enhancement in future phases.

2. Understanding of the Assignment

2.1 Business Context

DWSSM/SRIMS is at an early and critical stage of institutionalising service regulation as a formal government function. The Water Supply and Sanitation Regulations 2081 have created the legal basis for this function, but the institutional systems, data management practices, and reporting tools required to operationalise regulatory oversight are still being established. SRIMS currently manages information on 87 registered operators using manual spreadsheet workflows, which limits the timeliness of regulatory evidence and the ability to generate structured reporting for Development Partner coordination and government accountability.

Development Partners active in Nepal's water and sanitation sector make financial and technical commitments that are closely aligned with the regulatory implementation roadmap. At present, there is no single consolidated platform that allows SRIMS to track these commitments against regulatory milestones, monitor utilisation, and produce structured coordination reports. This limits both transparency and the efficiency of Development Partner engagement.

The RRPCD will provide SRIMS with a structured, institutionally owned data management environment that is appropriate to the current phase of regulatory development. Rather than imposing a complex enterprise system on an emerging function, the platform will be purpose-fitted to SRIMS's immediate operational needs while being designed for integration into the broader national WASH information architecture as those needs evolve.

2.2 Existing Challenges

Framgmented Operator Data Management

Performance data on the 87 registered WSSOs is currently maintained across spreadsheets, periodic sectoral reports, and institutional records held at different levels of the organisation. There is no consolidated repository enabling cross-operator performance comparison, trend analysis, or the generation of timely evidence for regulatory decisions. Data retrieval for reporting purposes requires significant manual effort and is subject to inconsistency across files.

Absence of a Regulatory Milestone Tracking Tool

The regulatory implementation roadmap — encompassing milestones related to licensing, KPI operationalisation, tariff review, compliance monitoring, and enforcement — is not currently tracked through any structured digital tool. Progress is assessed through periodic internal reviews and correspondence, which limits the ability to communicate implementation status to Development Partners, the Ministry, and other stakeholders.

Unstructured Development Partner Coordination

Multiple Development Partners are providing financial and technical support to the water and sanitation regulatory agenda. Their contributions are tracked through separate bilateral communications and reporting systems. There is no single view of DP engagement that maps contributions against regulatory milestones, making it difficult to identify coverage gaps, avoid duplication, and produce integrated coordination outputs.

Lack of Institutional Digital Capacity for Regulation

Service regulation as a formal institutional function is nascent in Nepal. SRIMS does not yet have dedicated digital infrastructure for regulatory information management, nor established workflows

for structured data collection from operators. The RRPCD must therefore be designed not only as a reporting tool but as a capacity-building platform that establishes foundational data management practices for ongoing regulatory oversight.

2.3 Expected Outcomes

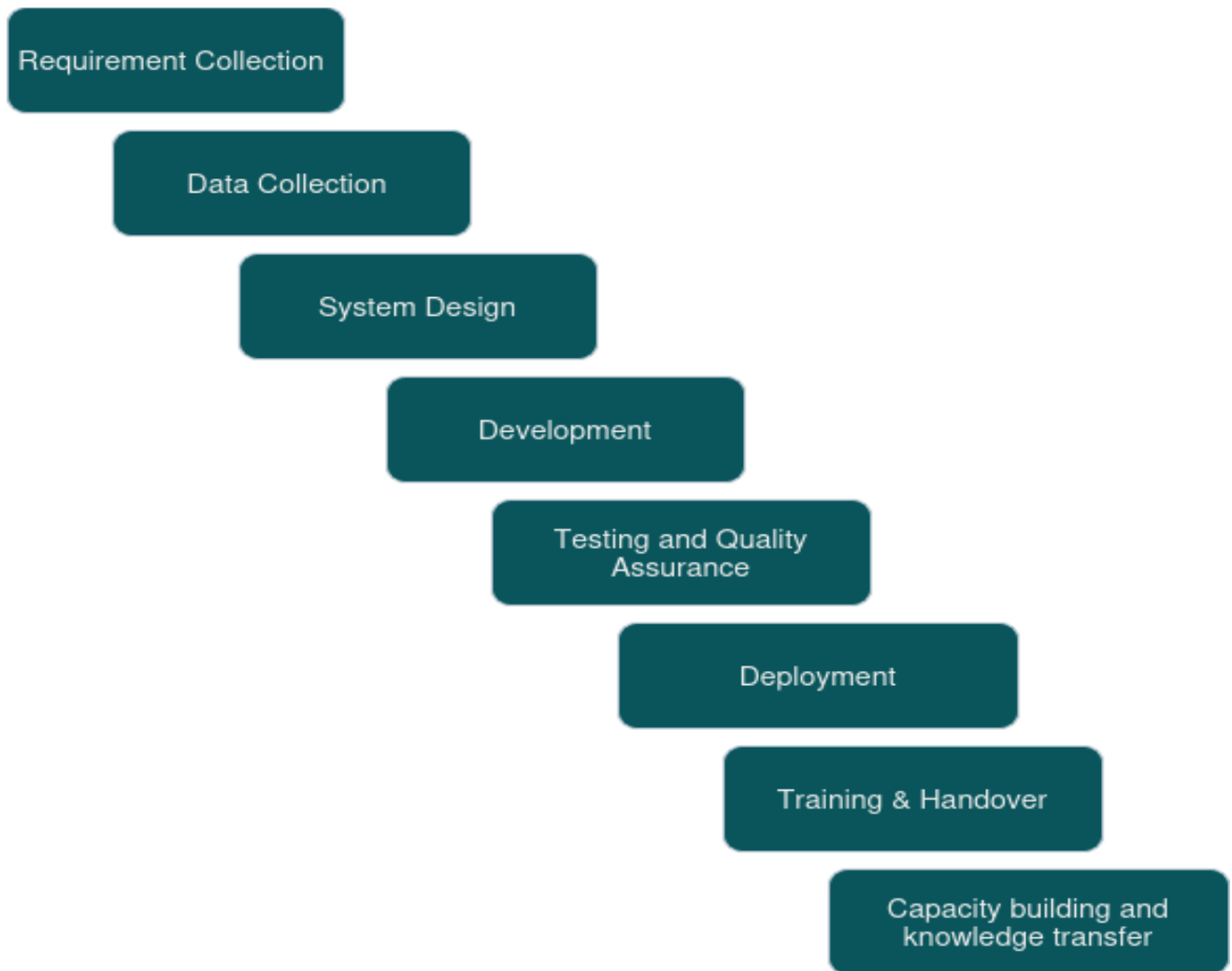
Upon successful completion of this assignment, the following outcomes are expected:

- A fully functional, deployed web-based dashboard providing SRIMS staff with consolidated visibility of regulatory milestones, operator KPI performance, and Development Partner engagement.
- A reduction in time and effort required to compile regulatory status updates and Development Partner coordination reports, through structured data access and dashboard-based reporting.
- A structured digital repository of operator performance data drawn from the existing SRIMS master dataset, forming the foundation for ongoing regulatory data management.
- A maintainable, documented platform that SRIMS non-technical staff can administer and update independently.
- A codebase and architecture designed to accommodate future integration with the National WASH MIS as regulatory functions and data infrastructure mature.

3. Proposed Methodology

The Consultant proposes an eight-phase delivery methodology designed to ensure structured, quality-controlled delivery within the 90-day assignment duration. The methodology is iterative: each phase produces defined outputs that are reviewed before the subsequent phase proceeds, ensuring alignment with DWSSM/SRIMS requirements at each stage. Training, handover, and capacity building are incorporated as the final two phases of the methodology, ensuring that knowledge transfer is treated as an integral delivery activity rather than an afterthought.

Methodology Flow Diagram:



3.1 Phase 1 — Requirement Collection

Purpose: To confirm the functional and technical requirements of the RRPCD through structured engagement with DWSSM/SRIMS focal persons, building directly on the TOR and the existing SRIMS master dataset.

Activities:

- Inception meeting with the designated SRIMS focal persons to confirm scope, data availability, and visual design direction.
- Structured review of the SRIMS master dataset to understand current data fields, operator records, KPI definitions, and update frequency.
- Review of the Water Supply and Sanitation Regulations 2081 and existing regulatory roadmap documentation to define the milestone structure for the roadmap module.

Expected Output: Confirmed requirements document covering module-level data structures, user roles, visual design preferences. This output will inform the system design phase.

3.2 Phase 2 — Data Collection

Purpose: To collect, assess, and structure all source data required to populate the dashboard, identifying data quality issues and establishing the canonical data schema for each module.

Activities:

- Receipt and review of the SRIMS master dataset (Excel format) containing operator profiles and KPI records.
- Mapping of existing dataset fields to the proposed dashboard data schema; identification of gaps, inconsistencies, and fields requiring standardisation.
- Collection of the current regulatory roadmap structure including phases, milestones, and target timelines.
- Collection of Development Partner profiles, engagement areas, and contribution data from SRIMS records.
- Definition of data validation rules and mandatory fields for each module based on dataset review findings.

Expected Output: Structured source datasets in agreed formats, a data schema specification, and a brief data assessment note documenting quality observations and proposed treatments.

3.3 Phase 3 — System Design

Purpose: To produce the complete technical design of the RRPCD prior to development, covering system architecture, database schema, UI/UX wireframes, and API specifications.

Activities:

- Development of the high-level and detailed system architecture covering frontend, backend, database, and integration layers.
- Design of the MongoDB database schema for all collections required by the four modules and the administrative backend.
- Design of wireframes for all screens within each module, presented to SRIMS for review and feedback.
- Definition of RESTful API endpoint specifications for data retrieval, administrative data entry, and bulk upload from Excel.
- Agreement on the visual design system including typography, colour palette, layout structure, and DWSSM/SRIMS branding elements.

Expected Output: System Design Document (SDD) covering architecture diagrams, database schema, wireframes, API specifications, and visual design system. This document must receive written approval from SRIMS before development commences.

3.4 Phase 4 — Development

Purpose: To build the RRPCD in iterative development cycles, with periodic demonstrations to SRIMS focal persons for review and feedback at key milestones.

Activities:

- Cycle 1: System foundation — navigation framework, base layout, database initialisation, and Excel data ingestion pipeline.
- Cycle 2: Dashboard Overview module and Regulatory Roadmap module with animated, interactive roadmap visualisation.
- Cycle 3: Operator Performance Module including operator profile cards, search and filter functionality, and KPI score display.
- Cycle 4: Development Partner Coordination Module, administrative data entry form, and GIS placeholder integration point.
- Periodic demonstrations to SRIMS focal persons at the end of each cycle for structured review and feedback.

Expected Output: Functional beta dashboard incorporating all four modules, submitted to DWSSM/SRIMS for structured review and consolidated feedback.

3.5 Phase 5 — Testing and Quality Assurance

Purpose: To verify that the beta dashboard meets all functional requirements, performs reliably, and is free of critical defects prior to deployment.

Activities:

- Functional testing of all modules against the agreed requirements specification, verifying data display, search and filter operations, and roadmap interactions.
- Cross-browser compatibility testing across current versions of Chrome, Firefox, Edge, and Safari.
- Responsive design testing across desktop and tablet viewport sizes.
- Data integrity testing to verify accuracy of KPI data loaded from the SRIMS master dataset.
- User Acceptance Testing (UAT) with designated SRIMS staff; defects logged, triaged, and resolved prior to final deployment.

Expected Output: All critical and major defects resolved.

3.6 Phase 6 — Deployment

Purpose: To deploy the finalised dashboard to the agreed hosting environment .

Activities:

- Finalisation of the production hosting environment based on client-confirmed infrastructure specifications.
- Execution of the deployment checklist covering database migration, environment configuration, smoke testing, and go-live validation.
- Configuration of administrative user accounts and access permissions for designated SRIMS staff.
- Preparation and finalisation of the User Manual and Technical Documentation in parallel with deployment activities.

Expected Output: Deployed production dashboard accessible to designated SRIMS users, with go-live confirmation issued to DWSSM/SRIMS.

3.7 Phase 7 — Training and Handover

Purpose: To transfer operational knowledge of the dashboard to SRIMS staff and formally hand over all project deliverables.

Activities:

- Delivery of a structured training session for SRIMS administrative staff covering dashboard navigation, data update procedures, operator record management, and basic troubleshooting.
- Walkthrough of the administrative data entry form and Excel data ingestion workflow.
- Handover of the User Manual, Technical Documentation, and source code repository to DWSSM/SRIMS.
- Formal handover meeting and demonstration session with designated DWSSM/SRIMS staff.

Expected Output: Completed training programme; signed handover acknowledgement; delivery of User Manual, Technical Documentation, and version-controlled source code repository.

3.8 Phase 8 — Capacity Building and Knowledge Transfer

Purpose: To ensure SRIMS staff are equipped to administer, update, and further enhance the platform beyond the project period, enabling effective ownership and utilization of the system.

Activities:

- Conduct practical coaching sessions on platform administration, including data updates, user account management, content management, and use of key system functionalities.
- Provide guidance on procedures for updating regulatory roadmap information, Development Partner profiles, and other dashboard content as programme requirements evolve.
- Brief SRIMS personnel on the system architecture and design considerations relevant to future NWASH-MIS integration to support informed planning and decision-making in subsequent phases.
- Share lessons learned during implementation and provide recommendations for future platform enhancements and feature expansion opportunities

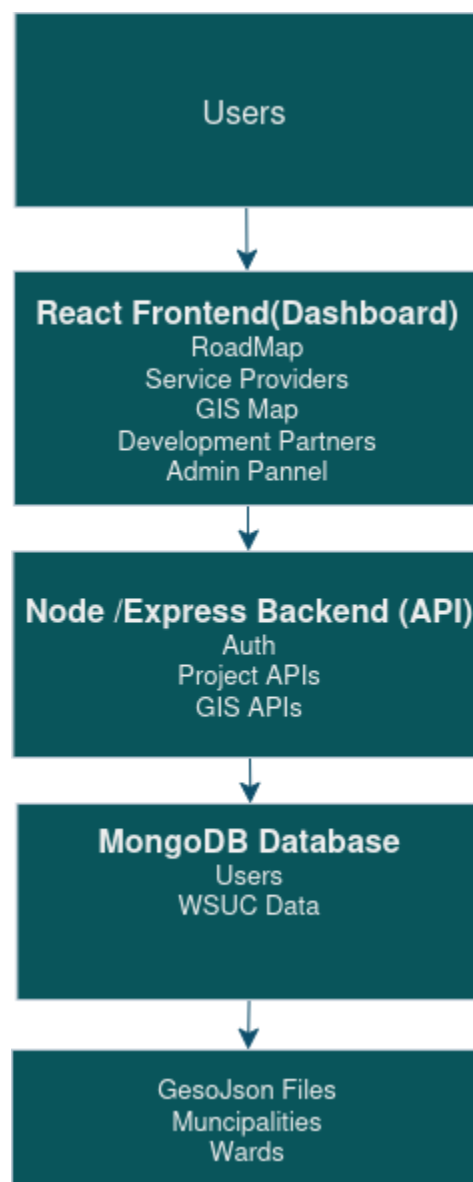
Expected Output: Capacity-building session completion record.

4. System Architecture

4.1 High-Level Architecture Overview

The RRPCD will be designed as a three-tier web application following client-server architecture. The presentation tier will be implemented as a single-page application (SPA) using React.js. The application tier will be implemented as a RESTful API server using Node.js and Express.js. The data tier will use MongoDB as the primary data store. All three tiers will be containerisable using Docker, enabling flexible deployment on cloud or on-premises infrastructure.

High-Level Architecture Diagram :



4.2 Frontend Architecture

The frontend application will be built with React.js using functional components and React Hooks for state management. A feature-based directory structure will organise each functional module as an independent unit containing its own components, hooks, API service calls, and utility functions. This structure ensures modularity and supports the independent development.

Styling will be implemented using Tailwind CSS, providing a utility-first approach that maintains visual consistency across all components. The design system will be aligned with DWSSM/SRIMS institutional branding, including the approved colour palette, typography, and logo usage. Client-side routing will be managed using React Router v6. Responsive design will be implemented across all views to support desktop and tablet access.

The animated, interactive Regulatory Roadmap will be implemented using React component state management and CSS animations, rendering the roadmap phases as an interactive visual timeline with clickable milestone nodes. The GIS integration point will be implemented using a Leaflet.js map component, configured to accept spatial data.

4.3 Backend Architecture

The backend API server will be implemented using Node.js with the Express.js framework, following RESTful design principles. The backend will implement a layered structure comprising: a routing layer for request parsing and delegation; a controller layer for business logic and response formatting; a service layer for reusable functions; a model layer defining MongoDB schemas using Mongoose; and a middleware layer for authentication, input validation, error handling, and rate limiting.

Authentication will be implemented using JSON Web Tokens (JWT). An Excel data ingestion service will process uploads of the SRIMS master dataset, parsing operator profiles and KPI records and mapping them to the dashboard data schema. A form-based data entry API will support direct record creation and updates by SRIMS administrative staff through the dashboard's administrative interface.

4.4 Database Architecture

MongoDB will serve as the primary data store. The schema design will follow a document-oriented approach reflecting the natural structure of regulatory data. The primary collections will be as follows:

Collection	Description
users	Authentication credentials, user roles, and access profiles for SRIMS staff and designated users
WSUC Data	Profiles of the 87 registered WSSOs including name, location, licence status, and contact details.

4.5 Integration Architecture

The RRPCD will include one integration components in this phase:

- Excel Data Integration: An upload interface will allow the SRIMS administrator to import the master dataset in Excel format. The integration service will parse the file, validate fields against the database schema, and populate the operators and kpi_records collections. An import log will record the results of each upload, flagging any records that fail validation.

Future integration with NWASH-MIS will be facilitated through RESTful API design patterns and standardised data formats. The data schema will document field-level alignment between the RRPCD and NWASH-MIS data models to support future integration planning.

5. Data Flow

5.1 Overview

The RRPCD handles data from two primary input channels: bulk data import from the SRIMS master Excel dataset, and direct data entry through the administrative form interface. Data flows out through the dashboard's visualisation components. The diagrams presented in Appendix A illustrate these flows at context level and process level.

5.2 Primary Data Sources

Data Source	Data Type	Update Method	Frequency
SRIMS Master Dataset (Excel)	Operator profiles, KPI scores, licensing status	Admin Excel upload via dashboard	Periodic (as SRIMS updates)
SRIMS Admin Form Input	Roadmap milestone updates, DP profile entries, status changes	Direct form entry via admin UI	As required
Regulatory Roadmap Records	Phase definitions, milestones, target and actual dates	Initial load + admin updates	As milestones evolve
Development Partner Records	DP profiles, commitments, engagement areas	Initial load + admin updates	As DP engagement evolves

5.3 Data Processing Flow

Upon receipt of data through either input channel, the following processing steps will be applied before data is committed to the MongoDB database:

1. Data ingestion: raw input received by the Express.js API layer and parsed into a structured object.
2. Schema validation: the parsed object validated against the Mongoose schema definition, checking data types, required fields, and format constraints.
3. Business rule validation: application-level rules applied as defined during the requirement collection phase (e.g., KPI score ranges, valid licence status values).
4. Duplicate detection: existing records checked for matching identifiers; update or reject applied according to the configured strategy.
5. Commit: validated records written to the relevant MongoDB collections. An audit log entry created recording the data source, timestamp, user identity, and validation outcome.

6. Functional Modules

The RRPCD will be structured around four functional modules, each addressing a specific regulatory information management need identified in the TOR. The following sections provide detailed specifications for each module.

6.1 Dashboard Overview Module

Purpose

The Dashboard Overview will serve as the primary landing page of the RRPCD. It will provide SRIMS staff with a consolidated, at-a-glance view of the regulatory programme's current status, presenting high-level KPI summaries, roadmap progress indicators, and key operator and DP contributions in a single accessible interface.

Key Features

- Summary KPI tiles displaying aggregate statistics: total registered operators, percentage of operators meeting KPI thresholds, active Development Partners.
- District-level visualizations showing operator distribution and compliance status across service areas
- Roadmap progress summary indicating current implementation phase and milestone status.
- Development Partner engagement summary listing active DPs and indicative total commitment figures.
- Navigation access to all four modules from a persistent sidebar and landing panel.

Inputs

- Aggregated operator and KPI data from the operators and kpi_records collections.
- Roadmap phase and milestone data from the roadmap_phases collection.
- Development Partner summary data from the dp_partners collection.

Outputs

- Rendered KPI tile panel with current values.
- Roadmap progress summary panel.
- Development Partner engagement summary panel.

6.2 Regulatory Roadmap Module

Purpose

The Regulatory Roadmap Module will provide an animated, interactive visualisation of DWSSM/SRIMS's regulatory implementation roadmap. It will enable staff and Development Partners to track progress against milestones, understand the sequencing of regulatory activities, and access milestone detail in a structured, transparent manner.

Key Features

- Statistics summary panel displaying total WSUCs, provinces covered, available SPI records, and service coverage readiness status.

- Service provider profile cards presenting key information including provider identity, geographic coverage, and operational status.
- Performance monitoring dashboard displaying key indicators such as Water Quality Score, Reliability Score, SPI, and OEI.
- Geographic coverage overview enabling users to understand the distribution of service providers across provinces, districts, and municipalities.
- Service coverage readiness indicators highlighting providers that meet predefined service coverage requirements.
- Search and filtering capabilities for efficient access to service provider information and performance records.
- Automated data synchronization ensuring the dashboard reflects the latest available service provider data.

Inputs

- Roadmap phase and milestone definitions from the roadmap_phases collection.
- Development Partner contribution data aligned to roadmap phases from the dp_partners collection.

Outputs

- Rendered interactive regulatory roadmap timeline.
- Milestone detail panels on selection.
- Development Partner contribution overlay aligned to roadmap phases.

6.3 Operator Performance Module (Service Providers)

Purpose

The Operator Performance Module provides structured visibility of Nepal's registered Water Supply and Sanitation User Committees (WSUCs) performance metrics and service coverage status. It enables SRIMS staff to assess individual operator performance, coverage readiness, and regulatory compliance through an intuitive card-based interface with real-time data synchronization.

Key Features

- Statistics summary panel displaying total WSUCs, provinces covered, available Service Performance Index (SPI) records, and service coverage readiness status.
- Service provider profile cards presenting key information including WSUC identification, geographic coverage, service coverage status, and operational performance indicators.
- Performance monitoring dashboard displaying key metrics such as Water Quality Score, Reliability Score, Service Performance Index (SPI), and Operational Efficiency Index (OEI).
- Service coverage readiness indicators highlighting providers that meet predefined service coverage requirements.
- Geographic coverage overview enabling users to understand the distribution of service providers across provinces, districts, and municipalities.
- Search and filtering capabilities to support efficient identification and review of service provider information and performance records.

- Automated data synchronization ensuring that service provider information and performance indicators remain current and accessible to users.

Inputs

Service provider records from the getServiceProviders() API including:

- WSUC_ID and WSUC_Name
- Location data (Province_Name, District_Name, Municipality_Name)
- Service_Coverage_Prerequisite status (Yes/No)
- Summary_Index metrics (Water_Quality_Score, Reliability_Score, SPI, OEI)
- Complete provider metadata for comprehensive profile display

Outputs

- Rendered statistics panel with 4 aggregate metrics
- Service provider card grid displaying all WSUCs with:
 - Complete location and identification information
 - Real-time performance metrics (Water Quality, Reliability, SPI, OEI)
 - Service coverage readiness status
 - Responsive visual layout with interactive hover states
- Dynamic provider count badge showing total records

6.4 Development Partner Coordination Module

Purpose

The Development Partner Coordination Module will provide a structured view of the Development Partners active in Nepal's regulatory agenda, displaying their profiles, areas of engagement, indicative financial and technical commitments, and the alignment of their contributions to specific phases of the regulatory roadmap. It will support SRIMS in managing DP relationships and producing integrated coordination outputs.

Key Features

- Development Partner directory displaying profiles of active DPs including organisational overview, areas of engagement, and contact information.
- Commitment summary panel for each DP displaying indicative financial commitments, technical assistance contributions, and the roadmap phases supported.
- Visual mapping of DP contributions against regulatory roadmap phases, enabling SRIMS to identify phases with strong DP support and those requiring additional engagement.
- DP engagement matrix providing a cross-tabulated view of DP contributions across roadmap phases.

Inputs

- Development Partner profile data from the dp_partners collection.
- Roadmap phase data from the roadmap_phases collection for the contribution alignment visualisation.

Outputs

- Rendered DP directory with individual profile views.
- Roadmap-phase contribution alignment visualisation.
- DP engagement matrix view.

6.5 Analytics & Charts Module: Operator Performance Categorization

Purpose

The Analytics & Charts Module provides advanced scatter plot-based visualization of water supply service provider performance through operational efficiency and service quality metrics. It enables SRIMS staff to categorize and analyze operators across nine performance zones, identify improvement trends, and filter providers by location and search criteria.

Key Features

- Interactive operator performance visualization using a scatter plot to assess service quality and operational efficiency across service providers.
- Performance categorization framework grouping operators into defined performance zones based on Operational Efficiency Index (OEI) and Service Performance Index (SPI) values.
- Comparative analysis capability enabling users to identify high-performing, improving, and underperforming service providers.
- Summary statistics panel presenting key indicators including total operators assessed, active performers, improved performers, and filtered results.
- Search and filtering functionality allows users to analyze operators by province, district, municipality, and service provider name.
- Interactive data exploration through tooltips and detailed operator information displayed within the visualization interface.
- Performance trend analysis supporting evidence-based decision-making and prioritization of regulatory interventions.
- Automated processing and visualization of operator performance data to ensure consistent and up-to-date analytical reporting.

Inputs

- Service provider records from getServiceProviders() API
- Service Performance Index (SPI) and Operational Efficiency Index (OEI) metrics
- Provider names (WSUC_Name) and IDs (WSUC_ID)
- Location data (Province_Name, District_Name, Municipality_Name)
- User search queries for provider name matching
- User-applied geographic filters

Outputs

- Interactive scatter plot with 9 performance zones
- Four summary metric tiles (Total Records, Improved Zone, Active Zone, Highlighted)
- Color-coded provider markers positioned by OEI and SPI values

- Animated point transitions on data updates
- Interactive tooltips with detailed provider metrics
- Highlighted search results with visual emphasis
- Searchable and filterable provider visualization

6.6 Administrative Backend Module

Purpose

The Administrative Backend Module will provide SRIMS non-technical staff with a simple, form-based interface to manage dashboard content without requiring direct database access or technical expertise. It will serve as the primary operational tool for maintaining data currency in the platform.

Key Features

- Excel upload interface for bulk import of operator and KPI records from the SRIMS master dataset, with import results and error reporting displayed on completion.
- Form-based editor for creating and updating individual operator profiles, KPI records, roadmap milestones, and Development Partner entries.
- User management panel for creating, editing, and deactivating system user accounts.
- Audit log viewer displaying a record of all data modifications for accountability and traceability purposes.

Inputs

- SRIMS master dataset (Excel file) uploaded by administrator.
- Form field inputs submitted by administrator for individual record management.

Outputs

- Updated database records in the operators, kpi_records, roadmap_phases, and dp_partners collections.
- Import result report displayed on the admin interface following each Excel upload.
- Audit log entries for all data modification actions.

6.6 GIS Dashboard Module

Purpose

The GIS Dashboard Module will provide an interactive geographic visualization of NWASH project coverage across Nepal. It will enable SRIMS staff and stakeholders to view project locations, understand geographic distribution, identify coverage areas, and navigate project information through an intuitive map-based interface.

Key Features

- Interactive Nepal map powered by Leaflet displaying all registered NWASH projects.
- Project location markers representing project implementation areas across municipalities.
- Project information popups displaying key project details upon marker selection.

- Ward boundary visualization highlighting wards covered by individual projects.
- Geographic filtering capabilities allowing users to filter projects by Province, District, Municipality, and Ward.
- Dynamic map updates reflecting selected geographic filters and project selections.
- Coverage visualization showing project distribution and areas of intervention.
- Map navigation controls including zoom, pan, and layer interaction.

Inputs

- Project records containing project location information, including Province, District, Municipality, and Wards Covered.
- Municipality and ward boundary datasets used for geographic visualization and spatial mapping.
- User-selected geographic filter criteria.

Outputs

- Interactive map displaying project locations across Nepal.
- Visualization of wards covered by selected projects.
- Filtered geographic view based on user-selected Province, District, Municipality, or Ward.
- Project information panels and map-based coverage insights supporting monitoring and decision-making.

7. Deliverables

The following deliverables will be submitted in accordance with the TOR. All reports and documents will be submitted in both editable electronic format and PDF. Source code will be delivered via a version-controlled repository (GitHub or GitLab) with appropriate access granted to DWSSM/SRIMS.

Re f.	Deliverable	Due	Format	Approver
D1	Inception Report	Day 7 from contract signing	PDF / DOCX	Section Chief, SRIMS
D2	Beta Dashboard	Day 20 from contract signing	Hosted demo / deployable package	SRIMS focal person(s)
D3	Final Dashboard	Day 60 from contract signing	Deployed system / deployable package	Section Chief, SRIMS
D4	User Manual	Day 70 from contract signing	PDF / DOCX	SRIMS focal person(s)
D5	Technical Documentation	Day 90 from contract signing	PDF / DOCX + source code repository	Section Chief, SRIMS

8. Work Plan and Timeline

Milestone	Description	Target Day
M1	Inception Report Submission	Day 7
M2	Requirements Confirmed & Data Collection Complete	Day 10
M3	System Design Document Completed	Day 13
M4	Development Cycle 1 Completed	Day 50
M5	Development Cycle 2 Completed	Day 62
M6	Beta Dashboard Demonstrated & Feedback Received	Day 72
M7	User Acceptance Testing & Refinement Completed	Day 79
M8	Final Dashboard Deployed & Go-Live Confirmed	Day 80
M9	Training, Documentation Handover & Project Closure	Day 90

Milestone	Activity	Start Day	End Day	Duration (Days)	Project Timeline (Days)														
					1	10	20	30	40	50	60	70	80	90					
M1	Inception Report	1	7	7															
M2	Requirements confirmed; data collection complete	8	10	3															
M3	System Design Document	11	13	3															
M4	Development Cycle 1 complete (system foundation, overview, roadmap)	14	50	37															
M5	Development Cycle 2 complete (operator module, DP module, GIS map module, admin backend module)	51	62	12															
M6	Beta dashboard demonstrated to SRIMS; consolidated feedback received	63	72	10															
M7	UAT, bug fixes, performance optimization & final validation	73	79	7															
M8	Final dashboard deployed; go-live confirmed	80	80	1															
M9	Training delivered; documentation handed over; project closed	81	90	10															

10. Conclusion

This Inception Report marks the formal commencement of the RRPCD development engagement under the contract between DWSSM/SRIMS and BreadFruit Pvt Ltd. It documents the Consultant's thorough understanding of the regulatory context, the institutional challenges motivating this assignment, and the specific technical and functional requirements that the RRPCD will be designed to address.

The proposed system — structured around four purposeful modules, an accessible administrative backend, and an architecture designed for future NWAASH-MIS compatibility — is calibrated to SRIMS's current institutional capacity and the 90-day assignment duration. The eight-phase methodology integrates training, handover, and capacity building as core delivery phases, ensuring that SRIMS staff are fully equipped to operate and maintain the platform independently at project closure.

The Consultant is confident that the proposed technical stack, the iterative development approach with periodic SRIMS review cycles, and the strong emphasis on documentation and knowledge transfer will result in a platform that delivers immediate operational value to SRIMS while establishing a sound foundation for future regulatory information management investment.

The Consultant respectfully requests that DWSSM/SRIMS review this Inception Report and provide written approval or documented revision requests within the five working day review period. Upon approval, the Consultant will proceed immediately to the Requirement Collection phase and schedule the inception meeting with SRIMS focal persons in accordance with the work plan presented in Section 8.

Submitted by

Breadfruit Private Limited

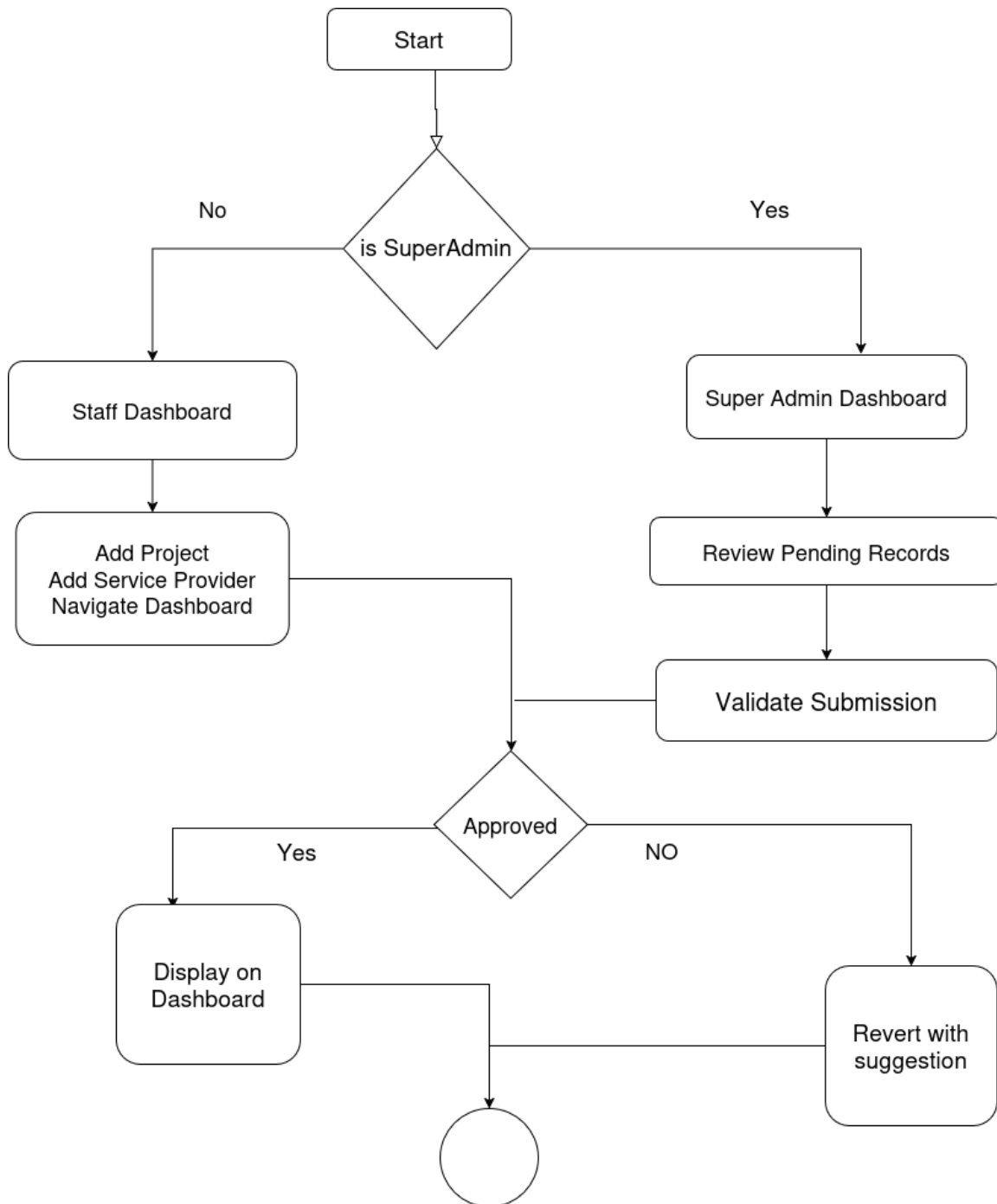
June 08, 2026



Appendix A — Flow Chart

The following text-based diagrams present the flow chart for the RRPCD.

[FIGURE A.1 — RRPCD: Flow Chart Diagram]



Appendix B — Glossary of Terms

Term / Abbreviation	Definition
DWSSM	Department of Water Supply and Sewerage Management. The Government of Nepal's technical department responsible for water supply and sanitation policy, planning, and regulation.
GIS	Geographic Information System. A system for capturing, storing, and presenting spatial or geographic data.
JWT	JSON Web Token. A compact means of representing identity claims between two parties, used in this system for user authentication.
KPI	Key Performance Indicator. A measurable value defined in the Water Supply and Sanitation Water Supply and Sanitation Regulations 2081 against which WSSO performance is assessed.
MongoDB	An open-source document-oriented database management system. Used as the primary data store for the RRPCD.
NWASH-MIS	National Water, Sanitation, and Hygiene Management Information System. The sector's existing central data platform serving local government service delivery monitoring.
RBAC	Role-Based Access Control. An approach to restricting system access based on the roles assigned to users.
React.js	An open-source JavaScript library for building user interfaces. Used as the frontend framework for the RRPCD.
RRPCD	Regulatory Roadmap and Partner Coordination Dashboard. The platform being developed under this engagement.
SPA	Single-Page Application. A web application that loads a single HTML page and dynamically updates content without full page reloads.
SRIMS	Service Regulation and Information Management Section. The DWSSM section responsible for regulatory oversight and information management, and the primary client for this engagement.
TOR	Terms of Reference. The document issued by DWSSM/SRIMS defining the scope, objectives, and requirements of this assignment.
UAT	User Acceptance Testing. A testing phase in which end users validate that a system meets their requirements before production deployment.
WSSO	Water Supply and Sanitation Operator. A registered service provider subject to regulatory oversight under the Water Supply and Sanitation Water Supply and Sanitation Regulations 2081. There are currently 87 registered WSSOs in Nepal.